# Two cheap rho jump tables compared: chained (main) against the ladder table (v5, v6)

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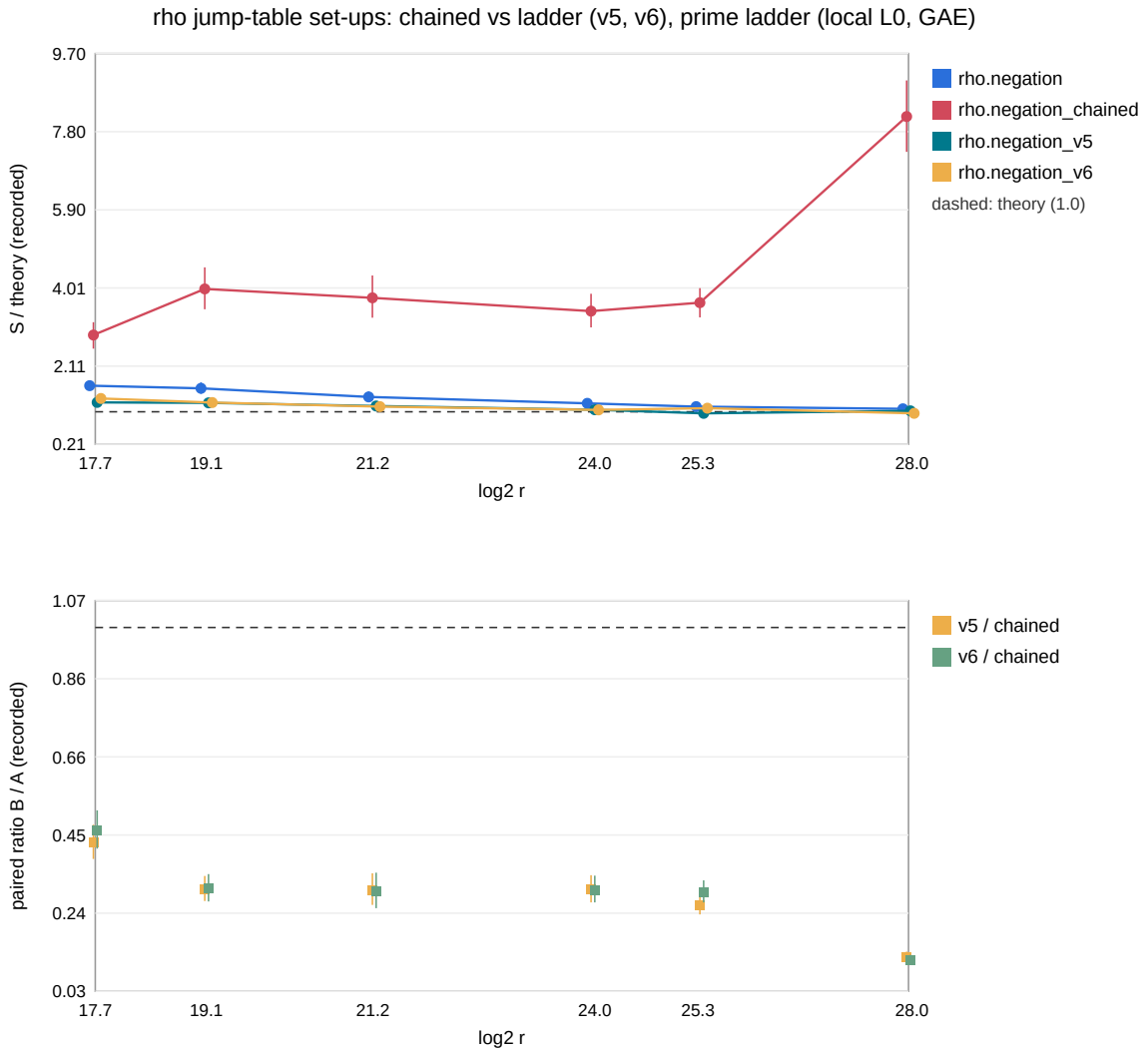
# Two cheap rho jump tables compared: chained against the ladder table (v5, v6)

**Result (measured, counted GAE, local L0 host) [pcc:unverified].** `rho.negation_chained` (from `research/curve25519_engine_items_20261110`) and this lab's `rho.negation_v5` / `rho.negation_v6` all build the negation-map rho jump table more cheaply than `rho.negation`. They were paired here on six prime curves, $2^{17.7–2}28$, with 32 targets × 6 rounds (1152 pairs per comparison) and an A/A control.

| comparison | paired ratio [95 %] | band (w = 0 / 0.1 / 1) |
|---|---|---|
| v5 / chained | **0.256 [0.236, 0.277]** | 0.256 / 0.257 / 0.261 |
| v6 / chained | **0.261 [0.240, 0.284]** | 0.261 / 0.263 / 0.269 |
| chained / rho.negation | 3.25 [2.98, 3.54] | 3.25 / 3.28 / 3.42 |
| v5 / rho.negation | 0.831 [0.797, 0.865] | 0.831 / 0.841 / 0.894 |
| v6 / rho.negation | 0.849 [0.814, 0.885] | 0.849 / 0.861 / 0.919 |
| chained A/A | 1.0000 | — |

Every order holds across the §5a band. The bounds (fit version 2, 6 sizes):

| bound | id | ops × floor | α |
|---|---|---|---|
| rho.negation_v5 | `ECBND1hbd448fdd4fba` | 1.103 | 0.461 |
| rho.negation_v6 | `ECBND1h61c3456f18f0` | 1.127 | 0.451 |
| rho.negation | `ECBND1h434d5b9180f8` | 1.328 | 0.426 |
| rho.negation_chained | `ECBND1ha265c33c87d3` | 4.312 | 0.597 |




S / theory per size (bound records) and the paired ratios against chained

# Why: the same set-up saving, opposite walks

The counters below are per √r, as recorded means:

| arm | set-up (adds + doublings) | walk operations | cycle escapes | look-ahead additions |
|---|---|---|---|---|
| rho.negation | 0.122 | 1.009 | 0.003 | 0.051 |
| rho.negation_chained | 0.051 | **3.708** | **0.045** | 0.167 |
| rho.negation_v5 | 0.046 | 0.884 | 0.000 | 0.011 |

Both cheap tables cut the set-up by the same amount, to about 0.05 √r. The chained table's jumps satisfy `M_j = M_{j−1} + M_{j−2}`, a three-term integer relation. Under the negation map that becomes a short fruitless cycle: cycles and escapes rise 15× and the walk runs 3.7× longer. The v5 table refuses every three-term relation among its jumps before spending an operation (`c = x ± y`, `c = y − x`, `c = 2x`, `2c = x`), and it uses k ≈ √F jumps instead of 4–16. Its walk comes to 0.88 √r, and cycles are rare: 270 of its 1,152 runs escaped one, with a mean of 0.0004 √r doublings, about 100× fewer than chained's 0.045 √r. The 0.000 in the table above is that value rounded.

This agrees in direction with the chained method's own study, which found 1.532× rho.negation at 20 bits and 2.738× at 24 bits and rejected it. Here the ratio is 3.25× pooled, and 8.2× theory at 2^28.

# Validation

An independent validator (`validation/`) recomputed every pooled ratio and band value to within rounding (v5 / chained 0.255689, chained / rho.negation 3.247918), and every mechanism mean (walk 3.70798 / 0.88383 / 1.00899). It re-checked [d]G = Q and recovered = planted on all 6,720 records. It corrected one sentence of the first version of this report, which said v5 “records no cycle” (see above).

It also found the band asymmetric. chained and rho.negation do not declare their distinguished-point table probes as unpriced work, while v5 and v6 do. Charging every arm’s table work at 1 GAE gives v5 / chained 0.258–0.260 (instead of 0.261) and chained / rho.negation 3.38–3.40 (instead of 3.42). No ranking changes at any weight. Per curve, chained / rho.negation rises from 1.75 at 2^17.7 to 7.62 at 2^28. The producer’s replay ran on the same host class, so it is not a cross-host replay.

# Run

| item | value |
|---|---|
| spec / session | `spec.json` / `ECBS1h56930a35f003`, 6720 records, all verified |
| commit / binary | `c7c4c2874f` (main, clean tree) / sha256 `55289451c555…` |
| audit | `audit.json`: replay-all 5760 reproduced, 0 problems |
| isolation | L0 container; counted GAE only |